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To the Editor-in-Chief
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Dear Editor,

We submit “Benign Exclusion and Consensus-Test Limits in Heterogeneous Federated Learning” for consideration as a regular paper.

The manuscript examines the cost that server-side defenses impose on honest clients under heterogeneous data and the decision paths behind observed failures. It combines adversary-free controls, matched random-selection experiments, and conditional analysis of our composite filter, Fed-MDBSCAN-G. The contribution is an empirical and mechanistic evaluation, without a claim of general superiority or a universal limitation of geometric defenses.

The historical evaluation covers eight server-side rules, four datasets, three Dirichlet concentrations, and five attack families with unbalanced coverage. Of 2,130 planned runs, 2,125 completed; five failures remain in the accounting. We additionally report 39 canonical re-executions and 36 control runs separately.

Matched random-selection controls show that admission count alone does not explain benign utility loss. At Dirichlet $\alpha = 0.01$, random selection of the same 61 updates improves accuracy over Multi-Krum by 15.9–43.7 percentage points on average across the tested datasets. For FLAME, random subsets matched to its recorded admission count in every round improve accuracy by 7.2–24.6 points on average, with improvement in every seed pair. The local and reference-library FLAME clustering selections agree in all 270 compared rounds; this does not establish equivalence of the authors’ complete training protocol.

For Fed-MDBSCAN-G, a sufficient radius condition prevents centroid validation from rejecting groups within the initial acceptance pool. It holds on 35 of 36 attacked five-step checkpoints and 22 of 72 attacked canonical checkpoints. Under the constrained probes, copied attackers are routed to the high-density side and bypass the cluster test. These findings distinguish geometric inactivity from routing failures and explain why added stages need not improve the final decision.

We believe this study fits TIFS through its examination of security evaluation for federated learning. The manuscript acknowledges prior findings on false positives under non-IID data and states the limits of its local implementations, three-seed evaluation, and retrospective analyses. It recommends reporting benign exclusion and utility alongside attack detection and attack success.

The submission includes the manuscript and supplementary material. Analysis scripts and summary evidence are available at <https://github.com/Bygokcen/fed-mdbscan-g>; reproducing historical checkpoint audits also requires the separate raw archive. Funding, competing interests, author contributions, and the use of AI-assisted tools are disclosed in the manuscript.

Sincerely,

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